

HW3 (ISL 6.6.5)

Tuesday, February 24, 2026

4:02 PM

(b) given: $X_{i1} = X_{i2}$; model becomes $\hat{y}_i = \beta_1 X_i + \beta_2 X_i = X_i(\beta_1 + \beta_2) = W X_i$

$$\beta_2 = W - \beta_1$$

If we use a placeholder $W = \beta_1 + \beta_2$

$$\text{we can rewrite: } \text{RSS} + \lambda(\beta_1^2 + \beta_2^2) = \sum (y_i - W X_i)^2 + \lambda(\beta_1^2 + (W - \beta_1)^2)$$

$$= \sum (y_i - W X_i)^2 + \lambda(2\beta_1^2 - 2W\beta_1 + W^2)$$

any sum of β_1 & β_2 values gives the same RSS (ridge penalty det. the coeff.)

we can disregard in exploration since RSS only depends on W & not indiv. values β_1 & β_2

$$\frac{d}{d\beta_1} : 4\beta_1 - 2W = 0 \quad \text{set}$$

substituting back in:

$$4\beta_1 - 2(\beta_1 + \beta_2) = 0 \rightarrow 4\beta_1 - 2\beta_1 - 2\beta_2 = 0$$

$$= 2\beta_1 - 2\beta_2 = 0 \rightarrow 2\beta_1 = 2\beta_2 \therefore \hat{\beta}_1 = \hat{\beta}_2$$

& since $W = \beta_1 + \beta_2$ we can state $W = 2\beta_1$

& so $\beta_1 = \beta_2 = \frac{W}{2}$ optimizes the ridge penalty

(d) ~ setup: $\text{RSS} + \lambda(|\beta_1| + |\beta_2|)$ w/ $W = \beta_1 + \beta_2$

once again the RSS term: $\sum (y_i - W X_i)^2$ only depends on W & not indiv. differences

EX: suppose the optimal sum of coeff has $W = 5$

RSS is not affected by changes in split btw β_1, β_2 whether it be $(\beta_1 = 1, \beta_2 = 4)$ or $(\beta_1 = 3, \beta_2 = 2)$

given those examples for Lasso w/ penalty: $|\beta_1| + |\beta_2|$

$$(\beta_1 = 1, \beta_2 = 4) \text{ follows } |1| + |4| = 5 \quad \& \quad (\beta_1 = 3, \beta_2 = 2) \text{ follows } |3| + |2| = 5$$

Therefore many solutions can optimize the Lasso penalty, e.g. an $\beta_1 + \beta_2 = W$

~ using this ex for Ridge: $\beta_1^2 + \beta_2^2$

$$(\beta_1 = 1, \beta_2 = 4) \text{ follows } 1^2 + 4^2 = 17 \quad \& \quad (\beta_1 = 3, \beta_2 = 2) \text{ follows } 3^2 + 2^2 = 13$$

~ we see $\beta_1^2 + \beta_2^2$ minimizes at $\beta_1 = \beta_2 = 2.5$ where $2.5^2 + 2.5^2 = 12.5$